Why zero-temperature SK minimizers have no support gaps

A short reconstruction of OpenAI's full-support argument · 23 October 2026

Abstract. For the pure Sherrington–Kirkpatrick model at zero external field, every admissible integrable minimizer of the zero-temperature Parisi functional has full relative Stieltjes support on $[0,1)$. This draft reconstructs the argument in OpenAI's September 2026 manuscript. Minimality gives a nonnegative contact potential. An integrated coercive estimate makes the derivative of a diffusion moment strictly convex on every positive constant interval of the order parameter. The resulting deficit between two contact points excludes an interior support gap; separate endpoint arguments complete the proof. We state the analytic input explicitly, derive the variational contradiction, and explain the resulting scalar value identity. The connection to the quantum approximate optimization algorithm (QAOA) requires further constructions beyond this support theorem.

1. The variational problem and the result

Let J_{ij} be independent standard Gaussian variables. The zero-field SK Hamiltonian is

$$H_n(\sigma) = \frac{1}{\sqrt{n}} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1,1\}^n.$$

We use the covariance convention $\xi(t) = t^2/2$. For configurations σ, τ with overlap $R = n^{-1}\sum_i \sigma_i\tau_i$, the covariance of this unordered-pair Hamiltonian is $nR^2/2 - 1/2$. Adding a single independent centered Gaussian of variance $1/2$, common to all configurations, gives covariance $n\xi(R)$ and does not change the expected maximum.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ with $\int_0^1 \gamma(t)\,dt < \infty$. Its Stieltjes measure is determined by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus $\mu_\gamma(\{0\}) = \gamma(0)$. This measure is locally finite; its total mass need not be finite and it is not normalized to be a probability measure. Support always means support in the relative topology of $[0,1)$.

For $\gamma \in \mathcal{U}$, let Φ_γ be the solution of

$$\partial_t\Phi_\gamma = -\frac{1}{2}(\partial_{xx}\Phi_\gamma + \gamma(t)(\partial_x\Phi_\gamma)^2), \qquad \Phi_\gamma(1,x) = |x|.$$

On compact time strips, the value function and its spatial derivatives are locally absolutely continuous in time, and the PDE holds almost everywhere in time. For nonsmooth coefficients the value function is defined by the stochastic-control formula, equivalently the monotone-coefficient approximation, in [FS, equation (1.1) and the regularity section]. Define

$$\mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \frac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

The zero-temperature Parisi formula and attainment identify $P_* := \lim_{n\to\infty} n^{-1}\mathbb{E}\max_\sigma H_n(\sigma)$ with $\min_{\gamma\in\mathcal{U}} \mathcal{P}(\gamma)$ [AC, Theorem 1 and the proof of Lemma 3]. The support theorem below is conditional on a minimizer; it does not itself prove existence.

Theorem 1 (full support; OpenAI [FS]). In the pure zero-field SK normalization $\xi(t) = t^2/2$, let $\gamma \in \mathcal{U}$ minimize $\mathcal{P}$. Then

$$\operatorname{supp} \mu_\gamma = [0, 1), \qquad \gamma(b) > \gamma(a) \quad (0 \le a < b < 1).$$

Writing $u = \partial_x \Phi_\gamma$, the associated diffusion

$$dX_t = \gamma(t) u(t, X_t)\, dt + dB_t, \qquad X_0 = 0,$$

exists, with B standard Brownian motion, and satisfies

$$\mathbb{E}[u(t, X_t)^2] = t, \qquad \mathbb{E}[(\partial_{xx} \Phi_\gamma(t, X_t))^2] = 1 \quad (0 \le t < 1).$$

Integrability allows γ to grow near one; the theorem does not assume a bounded or finitely stepped order parameter. Zero field supplies the symmetry $u(0, 0) = 0$, and the constant value $\xi'' = 1$ enters the curvature calculation. Extensions to nonzero field or mixed-spin covariance are not asserted here.

Infinitely many support points would not suffice. For example, $\sum_{k \ge 1} 2^{-k} \delta_{1-2^{-k}}$ has infinitely many support points accumulating at one and still leaves open intervals without mass. Theorem 1 excludes every such interval. Auffinger, Chen and Zeng proved infinite support [ACZ]. Chen's Theorem 1.3 gives the same full-relative-support conclusion, together with smooth absolute continuity [Chen]. The argument reconstructed here is OpenAI's explicit coercive-identity route to excluding gaps. This note claims neither a new support theorem nor priority for that conclusion.

2. Minimality creates contact points

Fix a minimizer and set

$$q(t) = \mathbb{E}[u(t, X_t)^2], \qquad G(s) = \int_s^1 (q(t) - t)\, dt.$$

The analytic regularity and diffusion results of [FS] give, on each compact time interval below one,

$$du(t, X_t) = \partial_{xx} \Phi_\gamma(t, X_t)\, dB_t, \qquad q'(t) = \mathbb{E}[(\partial_{xx} \Phi_\gamma(t, X_t))^2].$$

In particular $q(0) = 0$. The same source proves the one-sided variation formula

$$D\mathcal{P}(\gamma)[\eta] = \frac{1}{2} \int_0^1 \eta(t)(q(t) - t)\, dt$$

for each integrable direction η for which $\gamma + \varepsilon\eta \in \mathcal{U}$ for all sufficiently small $\varepsilon \ge 0$. We now extract its consequence without assuming that the Stieltjes measure has atoms or a density.

Adding an atom at s corresponds to $\eta(t) = \mathbf{1}_{[s,1)}(t)$. Minimality therefore gives $G(s) \ge 0$. For a compact $K \subset [0, 1)$, restrict μ_γ to K and write this finite measure as ν. Removing a fraction of this mass gives the admissible direction $\eta(t) = -\nu([0, t])$. Fubini's theorem yields

$$0 \le D\mathcal{P}(\gamma)[\eta] = -\frac{1}{2} \int_K G(s)\, d\mu_\gamma(s).$$

Since $G \ge 0$, it vanishes μ_γ-almost everywhere on every such K. Continuity then gives $G = 0$ on $S = \operatorname{supp} \mu_\gamma$.

At an interior support point s, the function G has a minimum. As $G' = s - q(s)$ and $G'' = 1 - q'$, we obtain

$$q(s) = s, \qquad q'(s) \le 1. \tag{1}$$

If $0 \in S$, then $G(0) = q(0) = 0$ and $G(h) = \frac{1}{2}(1 - q'(0))h^2 + o(h^2)$, so $q'(0) \le 1$ as well. These are contact conditions, not yet the identity $q(t) = t$ everywhere. Their proof uses the freedom to add and remove Stieltjes mass without a probability normalization constraint.

3. The estimate that excludes a gap

A gap in S is a constant interval of γ. The following analytic input is the hard step of [FS, Proposition 5.1].

Lemma 2 (constant-interval curvature). Let $\gamma \in \mathcal{U}$, without assuming minimality, and suppose that $\gamma = c$ on an open interval $I \subset (0, 1)$. If $c > 0$, then for every $t \in I$,

$$q'''(t) \ge \frac{1}{1000c^2}\mathbb{E}[(c\partial_{xx}\Phi_\gamma(t, X_t))^4] > 0. \tag{2}$$

Consequently q' is strictly convex on I. If $c = 0$, then q' is strictly increasing on I.

Here is the mechanism behind (2), with the analytic sign-propagation lemmas retained as cited inputs. On a positive constant interval, put

$$r = c\Phi_x, \qquad v = r_x, \qquad w = r_{xx}, \qquad z = r_{xxx}.$$

Differentiation along the diffusion gives

$$c^2 q'''(t) = \mathbb{E}[z^2 - 12vw^2 + 6v^4](t, X_t). \tag{3}$$

The negative middle term prevents a pointwise positivity argument. Even convexity of Φ does not settle the sign of this expression.

For a positive nondecreasing finite-step approximation $\tilde{\gamma}$ and terminal data $M^{-1}\log(2\cosh(Mx))$ with $M \ge \|\tilde{\gamma}\|_\infty$, the diffusion density factors as $f = e^{c\Phi}g$, where g obeys a forward heat equation between coefficient jumps. The source propagates backward inequalities for r and forward inequalities for the score $s = -\partial_x \log g$. On $x > 0$, r is increasing and concave; its backward heat-time derivative obeys $r_\tau / r \le 0$, with this ratio nondecreasing in x. The score s is increasing and concave, and s/r is nondecreasing in x. The ratios extend smoothly through zero. These are substantive analytic inputs to the certificate. The source then defines a polynomial differential expression $\mathcal{W}(F)$ satisfying

$$f\mathcal{W}(F) = \partial_x(rfF).$$

The exact rational identity in [FS, Lemma 4.1] rewrites the integrand of (3) as this derivative correction, weighted squares, and terms nonnegative under those propagated inequalities. The remainder contains a v^4 term with coefficient

$$\frac{1661507}{1521000000} > \frac{1}{1000}.$$

The proved tail and integrability bounds make the derivative integrate to zero. Therefore the identity gives

$$\mathbb{E}_f[z^2 - 12vw^2 + 6v^4] \ge \frac{1}{1000}\mathbb{E}_f[v^4]. \tag{4}$$

This is an **integrated** inequality under specific profile and density conditions, not a pointwise inequality in arbitrary v, w, z.

The source next approximates an integrable coefficient by positive nondecreasing finite-step coefficients that remain equal to c near the chosen time. Spatial derivatives of Φ through order four converge uniformly there, and the coupled diffusions converge. Thus (4) passes to the limit using only the derivative moments appearing in it. No limit of the auxiliary density scores or their ratios is required. Strict positivity of Φ_{xx} makes the right side positive. The case $c = 0$ follows separately from the heat equation; division by c does not address it.

The full sign propagation, polynomial identity and approximation proofs are in FS, `signs.tex`, `certificate.tex`, `curvature.tex`. The long reasoning record, §4, "Strict curvature on a constant interval", expands the mechanism and records the exact source destinations. This section explains that input rather than reproducing its coefficient table or claiming an independent verification of it.

4. Eliminating all support gaps

We can now complete the variational argument. Suppose first that (a, b) is a gap with $0 \le a < b < 1$ and both endpoints in S. The contact conditions give $q(a) = a$, $q(b) = b$ and $q'(a), q'(b) \le 1$, with the right derivative at zero when needed. If the constant value of γ is positive, strict convexity and continuity at the endpoints imply

$$q'(t) < \frac{b-t}{b-a} q'(a) + \frac{t-a}{b-a} q'(b) \le 1 \qquad (a < t < b).$$

Integrating gives $q(b) - q(a) < b - a$, contradicting contact at the two endpoints. If the constant is zero, strict increase gives $q'(t) < q'(b) \le 1$ and yields the same contradiction.

An initial gap $[0, b)$ with $0 \notin S$ has $\gamma = 0$ throughout: there is neither an atom at zero nor any later mass before b. Again $q'(t) < q'(b) \le 1$, whereas $q(0) = 0$ and $q(b) = b$. Integration excludes this gap.

A terminal gap requires another argument because its endpoint 1 is outside the support domain. On a hypothetical constant interval $(a, 1)$, [FS, Lemma 6.3] proves

$$1 - q(t) \ge C\sqrt{1-t}$$

for some $C > 0$ and all t sufficiently close to one. The estimate comes from the terminal cusp $|x|$: in a window $|x| \le \sqrt{1-t}$ the smoothed gradient stays uniformly away from both -1 and 1, while the diffusion density has a uniform positive lower bound. Integrating over that window gives the displayed square-root deficit. It follows that $q(t) - t \le (1-t) - C\sqrt{1-t} < 0$ near one, contradicting $G(s) \ge 0$ for s sufficiently close to one. This also excludes empty support, when $\gamma = 0$ everywhere. The terminal-plateau conclusion was already recorded in [ACZ, Remark 7].

The relatively closed set S has no component in its complement, so $S = [0, 1)$. Equation (1) now holds at every positive time, and symmetry supplies $q(0) = 0$. Differentiating $q(t) = t$ gives the second moment identity in Theorem 1, including its one-sided value at zero. Finally, every nonempty open interval has positive μ_γ-mass, whence $\gamma(b) - \gamma(a) = \mu_\gamma((a, b]) > 0$. This proves Theorem 1 using the analytic input stated above. The long reasoning record, §5, "Excluding terminal, interior and initial gaps", retains the case separation and source destinations.

5. A scalar consequence and the QAOA question

The moment identities explain why full support matters to the value problem. Let $M_t = u(t, X_t)$ and $a(t,x) = \Phi_{xx}(t,x)$. Then $M_t = \int_0^t a(s, X_s)\, dB_s$ is a bounded martingale with $\mathbb{E}M_t^2 = t$. Its terminal limit U_1 satisfies $|U_1| \le 1$ and $\mathbb{E}U_1^2 = 1$, hence $U_1^2 = 1$ almost surely. The terminal convergence in [FS] gives $U_1 X_1 = |X_1|$ without requiring that $X_1 \neq 0$ almost surely.

Itô's formula and $q(t) = t$ give

$$\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt.$$

Multiplying $X_1 = B_1 + \int_0^1 \gamma(t) M_t\, dt$ by U_1, taking expectations, and using $\mathbb{E}[U_1 M_t] = \mathbb{E}M_t^2 = t$ then yields

$$P_* = \mathbb{E}[U_1 B_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{5}$$

The use of P_* here invokes the Parisi formula [AC]; the preceding identity is already an identity for $\mathcal{P}(\gamma)$. The integral is finite since $\mathbb{E}|a(t,X_t)| \le (\mathbb{E}a(t,X_t)^2)^{1/2} = 1$. This is [FS, Corollary 7.1]; the long reasoning record, §6, "Recovering the value from a martingale", provides the terminal and stochastic-integral justification.

The companion finite Gaussian construction approximates the right side of (5) first on $[0,T]$, with fixed $0 < T < 1$, and then as $T \uparrow 1$. Its mesh count tends to infinity before T approaches one. Those coefficients are scalar Gaussian correlations, not circuit energies. The approximation is treated in [FS, Proposition 7.2] and the long reasoning record, §7, "A finite Gaussian spin and its adjacent correlations".

OpenAI's separate QAOA manuscript [Q] targets a further statement. For a p-layer circuit with deterministic cost and mixer angles θ, β, let $v_p(\theta, \beta)$ be its limiting expected energy per spin as $n \to \infty$, holding those angles fixed. Define

$$Q_p = \sup_{\theta,\beta \in \mathbb{R}^p} v_p(\theta, \beta).$$

The manuscript's main theorem asserts $\lim_{p\to\infty} Q_p = P_*$: for every $\varepsilon > 0$, a finite depth and deterministic angles, independent of both n and the disorder, attain at least $P_* - \varepsilon$ in that limiting expectation. The angles and depth may depend on ε. No quantitative depth bound, finite-size estimate or efficient angle-selection procedure is supplied by that statement.

The zero-temperature route uses scalar value approximation, a classical synthesis construction, temporary longitudinal seed controls and their removal, and a final error assembly. The support theorem supplies only an earlier ingredient in that route. The manuscript also gives an independent positive-temperature route. Its full QAOA argument is outside this note; see [Q, sections 3–6 and Appendix A] and the long reasoning record, §9, "From the scalar formula to one ordinary QAOA word".

6. Source and verification scope

This is an AI-assisted draft reconstruction of the cited public arguments, not a new result or an independent verification claim. OpenAI is the credited author of [FS] and [Q]. The exact source edition is `openai/math@adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The accompanying evidence graph records source pins and check status; the passage correspondence record binds this text to that graph and the source locators.

The selected full-support interface takes an admissible minimizer as input and includes Stieltjes-measure and diffusion existence, full relative support, strict increase and the two moment identities. In the companion scalar-value interface, scalar and finite-Gaussian conclusions require a minimizer and an associated diffusion; its final ground-state/Parisi equality requires the minimizer premise alone. The source example records no fresh Lean build, axiom audit or Comparator replay of these targets. The admitted challenge statements and the implementing source files are different objects. Neither selected declaration asserts QAOA circuit convergence. Source inspection and the exposition checks for this draft do not constitute independent human review or peer review.

References

- **[FS]** OpenAI. *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter.* 27 September 2026. Exact manuscript and TeX source directory. Principal locators: Theorem 1.1; Lemmas 6.2 and 6.3; Proposition 5.1; Corollary 7.1.
- **[Q]** OpenAI. *QAOA attains the SK ground-state energy in the thermodynamic-first limit.* 25 September 2026. Exact manuscript. Theorem 1.1; construction in sections 3–6.
- **[AC]** Antonio Auffinger and Wei-Kuo Chen. "Parisi formula for the ground state energy in the mixed p-spin model." *Annals of Probability* **45** (2017), 4617–4631. Preprint. External formula and attainment input used in [FS].
- **[ACZ]** Antonio Auffinger, Wei-Kuo Chen and Qiang Zeng. "The SK model is infinite step replica symmetry breaking at zero temperature." *Communications on Pure and Applied Mathematics* **73** (2020), 921–943. Preprint. Infinite support and the terminal-plateau antecedent; these do not by themselves exclude an interior gap.
- **[Chen]** Hong-Bin Chen. *FRSB in the SK spin glass: convergence to full-interval support at zero temperature.* 20 July 2026, version 1. Preprint. Theorem 1.3 is credited by [FS] with the same full-support conclusion and smooth absolute continuity.